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UGC - Autonomous Institute
Accredited by NBA & NAAC with 'A' Grade
Approved by AICTE
Permanently affiliated to JNTUH

PROJECT BASED LEARNING

ON

INDUSTRIAL SHED STRUCTURAL DRAWING USING SOFTWARE

BY

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UNDER THE GUIDANCE OF

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NARSIMHA REDDY ENGINEERING COLLEGE (UGC-AUTONOMOUS)

APPROVED BY AICTE, AFFILIATED TO JNTUH, ACCREDITED BY NBA & NAAC WITH A- GRADE)

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500100.(2026-2027)

ABSTRACT

Industrial sheds are widely used for factories, warehouses, workshops, and storage facilities. They are generally constructed using steel structures, durability, and ease of construction. AUTOCAD is an important drafting software used to prepare accurate structural drawings such as layout plans, elevations, frame sections, foundation layouts, and steel connection details. This project demonstrates the structural detailing of a steel industry shed using AUTOCAD. It includes the preparation of a typical frame section with complete dimensions, member specifications, the project follows standard engineering practices and relevant Indian standard (IS codes), making it suitable for academic and practical applications.

1. INTRODUCTION

An industry shed is a large-span steel structure designed for industrial operations, manufacturing, storage, and commercial purposes. Steel sheds are preferred because they are economical, lightweight, and can cover large areas without intermediate columns.

AUTOCAD is widely used in civil and structural engineering to prepare professional engineering drawings with accurate dimensions, structural member layout, and detailing. proper structural drawings reduce construction error, improve communication between engineers and contractors, and ensure safe construction.

This project presents the complete structural drawing of a steel industrial shed, including plan, elevation, and sectional details.

2. DRAWING-1

Industrial shed layout plan overall dimensions

Overall dimensions

Item value

Shed length 25000mm

Shed width 20000mm

Grid spacing 5000mm

Number of bays 5

Number of frames 6

Plan includes

- Grid lines (A-E and 1-6)
- Steel column location
- Foundation position
- Roof bracing layout
- Centre lines
- Overall dimensions

3. DRAWING-2

Front elevation

Structural dimensions

Item value

Eave height 8000mm

Ridge height 10300mm

Roof slope 1:10

Column height 8000mm

Bay width 5000mm

Elevation includes

- Steel column
- Rafter
- Roof purlins
- Ridge ventilator
- Roof sheeting
- Foundation level
- Ground level
- Overall dimension

4. DRAWING-3

TYPICAL FRAME SECTION

Structural members

Member section

Steel column ISMB 250

Steel rafter ISMB 250

Purlin C150*50*20*2mm

Girt C150*50*20*2mm

Roof bracing ISA 60*60*6mm

Foundation details

Component size

RCC footing 1500*1500*450mm

Pedestal 450*450*600mm

PCC 100mm thick

Sectional includes

- Steel column
- Rafter
- Roof slope
- Base plate
- Anchor bolts
- RCC footing
- Ground level
- Dimensions

5. DRAWING DETAILING WITH MEASUREMENTS

Steel column Building dimensions

- Length=25000mm
- Width=20000mm
- Eave height=8000mm
- Ridge height=10300mm

Structural members

Member size

ISMB 250

Steel rafter ISMB 250

Purlin C150*50*20*2mm

Girt C150*50*20*2mm

Roof bracing ISA 60*60*6mm

FOUNDATION ITEM SIZE

Footing 1500*1500*450mm

PCC 100mm thick

Pedestal 450*450*600mm

Roof

- Roof slope=1:10
- Roofing sheet thickness=0.50mm GI sheets

6. AUTOCAD DRAWING TOOLS USED

DRAWING TOOLS

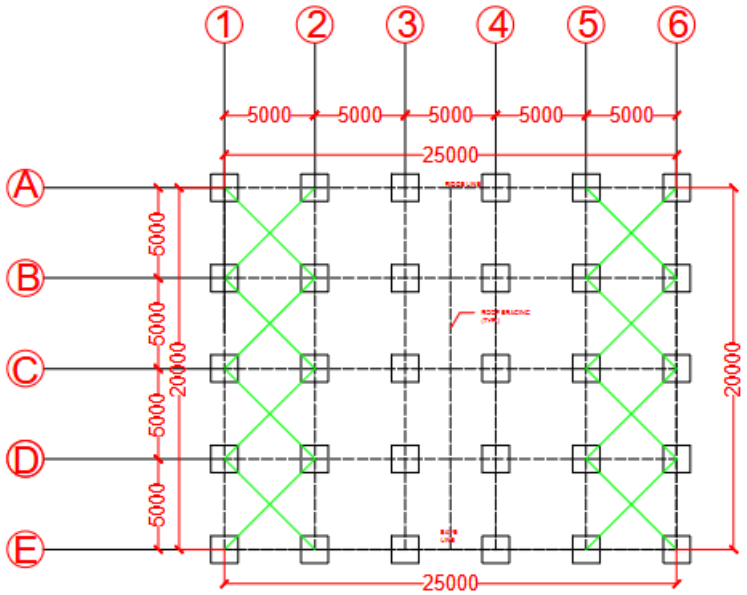
- LINE
- POLYLINE
- RECTANGLE
- CIRCLR
- ARC
- POLYGON
- HATCH
- TEXT
- MTEXT
- DIMENSION
- LEADER
- TABLE

LAYER TOOLS

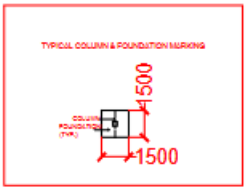
- LAYER PROPERTIES
- LINE TYPE
- LINE WEIGHT
- COLOUR MANAGEME

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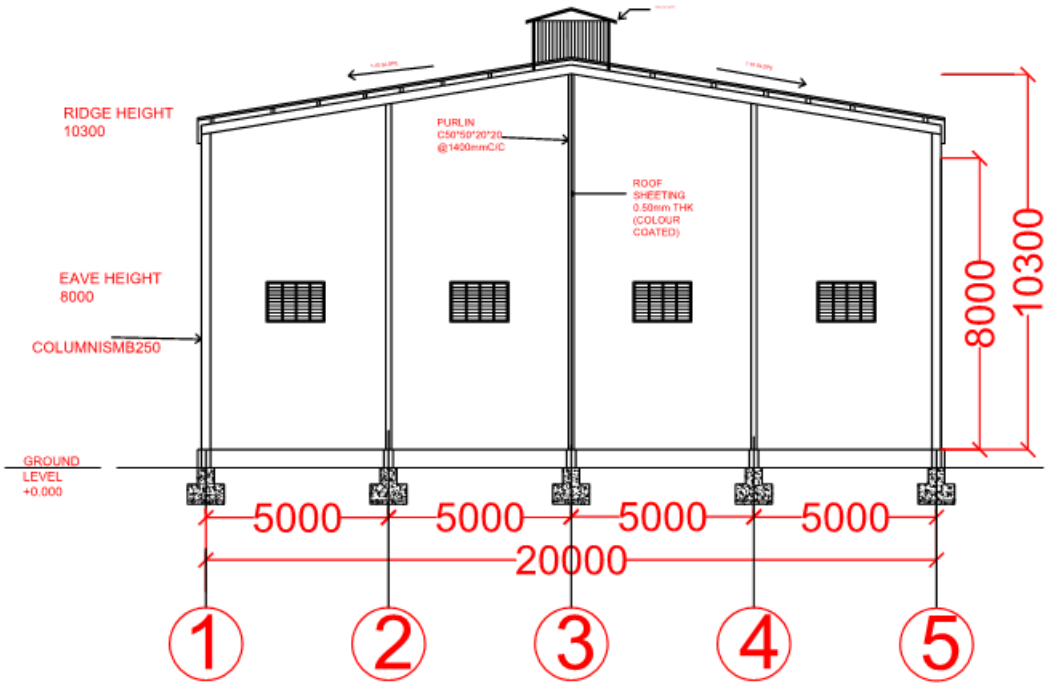
SHEET 1 - LAYOUT PLAN



PLAN



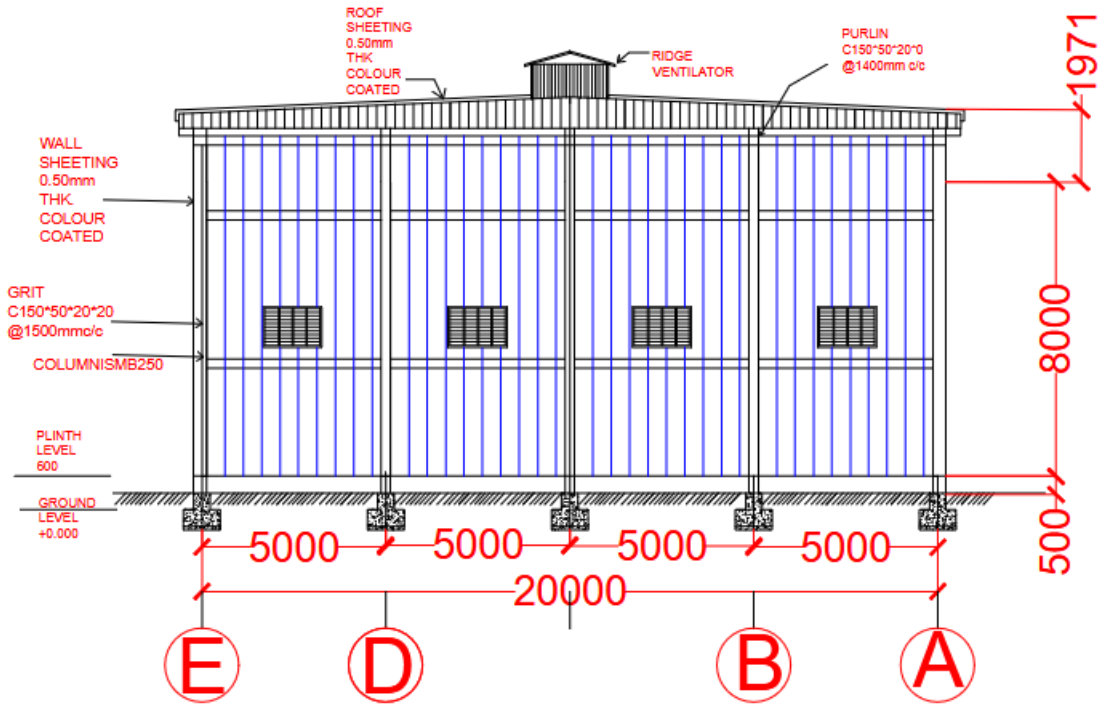
SHEET 2 -FRONT ELEVATION



FRONT ELEVATION (ALONG GRID 1)

WINDOW SCHEDULE				
TYPE	SIZE(mm)	SILL LEVEL	NO'S	REMARKS
W1	1500*1000mm	1200mm	4	LOUVRED WINDOW

SHEET 3 - SIDE ELEVATION (ALONG GRID E)



7. MODIFY TOOLS USED

- MOVE
- COPY
- MIRROR
- OFFSET
- TRIM
- EXTEND
- ROTATE
- SCALE
- STRETCH
- ARRAY
- FILLET
- CHAMFER
- BREAK
- EXPLODE
- JOIN

8.TECHNOLOGY USED SOFTWARE

- AUTOCAD 2024 operating system
- Windows 10/windows 11 design standard
- IS 800:2007(general construction in steel)
- IS 875 (LOADING STANDARDS)
- IS 456:2000 (concrete)
- National building code (NBC) 2016 hardware
- Intel core i5 processor or above
- 8 GB RAM
- 512 GB SSD
- Dedicated graphics card (recommended)

9. APPLICATIONS

- INDUSTRIAL FACTORIES
- WAREHOUSES
- MANUFACTURING PLANTS
- AUTOMOBILE WORKSHOPS
- AGRICULTURE STORAGE SHEDS
- LOGISTICS AND DISTRIBUTION CENTER
- AIRCRAFT HANGERS
- COMMERCIAL STORAGE BUILDINGS

10. ADVANTAGES

- Large column-free working spaces
- Faster construction
- High structural strength
- Lightweight steel structure
- Easy future expansion
- Accurate drafting using AUTOCAD
- Reduced construction errors
- Professional engineering documentation

11. CONCLUSION

The industrial shed structural drawing using AUTOCAD project demonstrates the preparation of accurate structural drawings for a steel industry shed. The project includes the layout plan, front elevation, and typical frame section with complete dimensions, member specification, and foundation details. AUTOCAD improves drafting accuracy, reduces construction errors, and provides professional engineering documentation. This project helps civil engineering students understand industrial steel structure and structural detailing techniques used in real-world construction projects.